[ECHELON PAVING]

When two or more pavers are placing bituminous material at the same time in the same direction on a roadway, that is commonly called echelon, or staggered, paving. Echelon paving creates a hot longitudinal joint for the compaction process and is a proven method to improve joint density. To perform echelon paving, it may be necessary to arrange for two asphalt plants to supply the material needed.

Note: If more than one plant is supplying material for the project, designate which paver is supplied by which plant.

Depending on the type of roadway, it may be necessary to close the roadway and direct traffic to alternate routes or set up traffic control to divert traffic to opposite lanes. Echelon paving also requires some additional planning.

- Calculate working speeds that balance the production rates of the pavers.
- Adjust working speeds to the capability of the least productive paver.
- Select pavers with similar screeds that develop uniformly high pre-compaction.
- When using non-similar screeds, adjust to the results produced by the lowest performing screed.
- Whenever possible, create similar paving widths for all pavers.
- When paving non-similar widths, paver with largest width should lead.
- Select compactors with drum widths that will create uniform rolling patterns behind each paver regardless of paving widths.



Assume that the base layer is being paved on a roadway that is 11.6 m (38') wide. The paving depth is 75 mm (3"). Two asphalt plants will be supplying the project with 400 tonnes (440 tons) per hour. The plan calls for three pavers to be used for the project.

The left side paver will pave 4.7 m (15.5') wide. That paving width includes an inside paved shoulder with 2 percent slope from right to left and the left driving lane with 2 percent slope from left to right.

The center paver will pave the right driving lane at a width of 3.7 m (12') and a 2 percent slope from left to right.

The right paver will pave the right emergency lane at a width of 3.2 m (10.5') at a slope of 4 percent from left to right.

The left paver will be slightly ahead of the other two pavers. Limit the longitudinal gap between pavers to no more than one paver length in order to maintain nearly equal temperatures between the layers, so density produced by the compactors across the width of paving is equal.

FIRST LAYER

To calculate the paving speeds of the three pavers, first add the three paving widths.

Paver 1: **4.7 m (15.5')**
Paver 2: **3.7 m (12')**
Paver 3: **3.2 m (10.5')**
Total width: **11.6 m (38')**

Now divide the total width by the width of each pass. The result will be the percentage of hourly production for each paver.

$$\text{Paver 1: } \frac{4.7\text{ m (15.5')}}{11.6\text{ m (38')}} = 40\%$$

$$\text{Paver 2: } \frac{3.7\text{ m (12')}}{11.6\text{ m (38')}} = 32\%$$

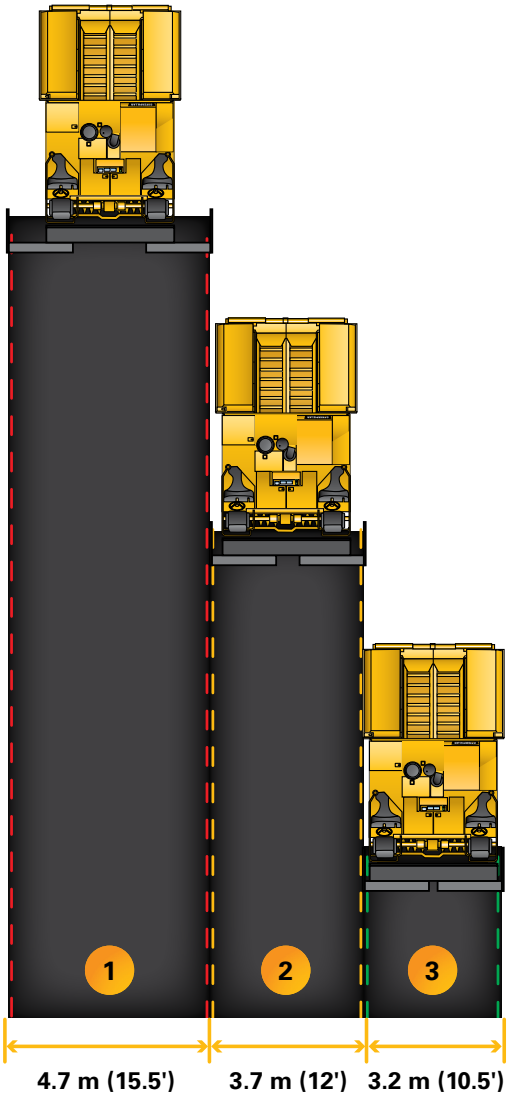
$$\text{Paver 3: } \frac{3.2\text{ m (10.5')}}{11.6\text{ m (38')}} = 28\%$$

Finally, divide the hourly production rate by the percentage of each paver. This will provide the hourly production rate in tons for each paver.

$$\text{Paver 1: } \frac{400\,\text{T}}{40\%} = \text{160 tonnes (176 tons)/hour}$$

$$\text{Paver 2: } \frac{400\,\text{T}}{32\%} = \text{128 tonnes (140 tons)/hour}$$

$$\text{Paver 3: } \frac{400\,\text{T}}{28\%} = \text{112 tonnes (124 tons)/hour}$$



paving speed: **3.7 m (12')/min.**
paving depth: **75 mm (3")**

The Paving Production Calculator can be used to confirm that each paver will be operating at the same speed based on the hourly tonnage allotted. Assume that material transfer vehicles will be used and the efficiency rate will be 95%.

PAVER ONE

		Imperial Units	Metric Units
Trucking	**General Inputs**		
Paver Speed	Paving Thickness:	[2.95] *in*	[75.0] *mm*
Compaction	Paving Width:	[15.42] *feet*	[4.7] *meter*
Windrow	Material Density Uncompacted:	[140] *lbs/ft³*	[2243] *kg/m³*
Yield	**Paver Speed @ Given Production Rate**		
Slope	Production Rate of Hot Plant:	[176] *tons/hr*	[160] *tonnes/hr*
Thickness	Calculated Paving Speed - 100% Efficiency:	[11.1] *ft/min*	[3.39] *m/min*
Job Summary	Calculated Paving Speed - 95% Efficiency:	[11.7] *ft/min*	[3.56] *m/min*
Legal	Calculated Paving Speed - 90% Efficiency:	[12.2] *ft/min*	[3.73] *m/min*
	Calculated Paving Speed - 85% Efficiency:	[12.8] *ft/min*	[3.90] *m/min*
	Calculated Paving Speed - 80% Efficiency:	[13.3] *ft/min*	[4.07] *m/min*
	Calculated Paving Speed - 75% Efficiency:	[13.9] *ft/min*	[4.24] *m/min*
Exit	**Effective Paving Speed**:	[11.1] *ft/min*	[3.39] *m/min*

The actual paving speed for Paver One is 3.6 m (12') per minute using 95 percent efficiency rate.

PAVER TWO

		Imperial Units	Metric Units
Trucking	**General Inputs**		
Paver Speed	Paving Thickness:	[2.95] *in*	[75.0] *mm*
Compaction	Paving Width:	[12.14] *feet*	[3.7] *meter*
Windrow	Material Density Uncompacted:	[140] *lbs/ft³*	[2243] *kg/m³*
Yield	**Paver Speed @ Given Production Rate**		
Slope	Production Rate of Hot Plant:	[141] *tons/hr*	[128] *tonnes/hr*
Thickness	Calculated Paving Speed - 100% Efficiency:	[11.2] *ft/min*	[3.42] *m/min*
Job Summary	Calculated Paving Speed - 95% Efficiency:	[11.8] *ft/min*	[3.59] *m/min*
Legal	Calculated Paving Speed - 90% Efficiency:	[12.3] *ft/min*	[3.76] *m/min*
	Calculated Paving Speed - 85% Efficiency:	[12.9] *ft/min*	[3.93] *m/min*
	Calculated Paving Speed - 80% Efficiency:	[13.4] *ft/min*	[4.10] *m/min*
	Calculated Paving Speed - 75% Efficiency:	[14.0] *ft/min*	[4.28] *m/min*
Exit	**Effective Paving Speed**:	[11.2] *ft/min*	[3.42] *m/min*

Paver Two operates at 3.7 meters (12') per minute also. Note that the hourly tonnage is less than Paver One and the paving width is also less than the Paver One.

PAVER THREE

- Trucking
- **Paver Speed**
- Compaction
- Windrow
- Yield
- Slope
- Thickness
- Job Summary
- Legal

- Exit

	Imperial Units	Metric Units
General Inputs		
Paving Thickness:	[2.95] *in*	[75.0] *mm*
Paving Width:	[10.50] *feet*	[3.2] *meter*
Material Density Uncompacted:	[140] *lbs/ft³*	[2243] *kg/m³*
Paver Speed @ Given Production Rate		
Production Rate of Hot Plant:	[123] *tons/hr*	[128] *tonnes/hr*
Calculated Paving Speed - 100% Efficiency:	[11.3] *ft/min*	[3.45] *m/min*
Calculated Paving Speed - 95% Efficiency:	[11.9] *ft/min*	[3.62] *m/min*
Calculated Paving Speed - 90% Efficiency:	[12.4] *ft/min*	[3.80] *m/min*
Calculated Paving Speed - 85% Efficiency:	[13.0] *ft/min*	[3.97] *m/min*
Calculated Paving Speed - 80% Efficiency:	[13.6] *ft/min*	[4.14] *m/min*
Calculated Paving Speed - 75% Efficiency:	[14.1] *ft/min*	[4.31] *m/min*
Effective Paving Speed:	[11.3] *ft/min*	[3.45] *m/min*

The paving speed calculation for Paver Three confirms that all three pavers are in balance, each with its own production rate based on the width of its pass.

The truck guides on the project are responsible for directing the haul units to the appropriate pavers and transfer vehicles. In some cases, one transfer vehicle may be supplying two pavers. Using one transfer vehicle to supply two pavers takes some coordination, but if the pavers have large hopper inserts and paving speeds under 6 m (20') per minute, it can be done without stopping the pavers.

The next layer will require new calculations.

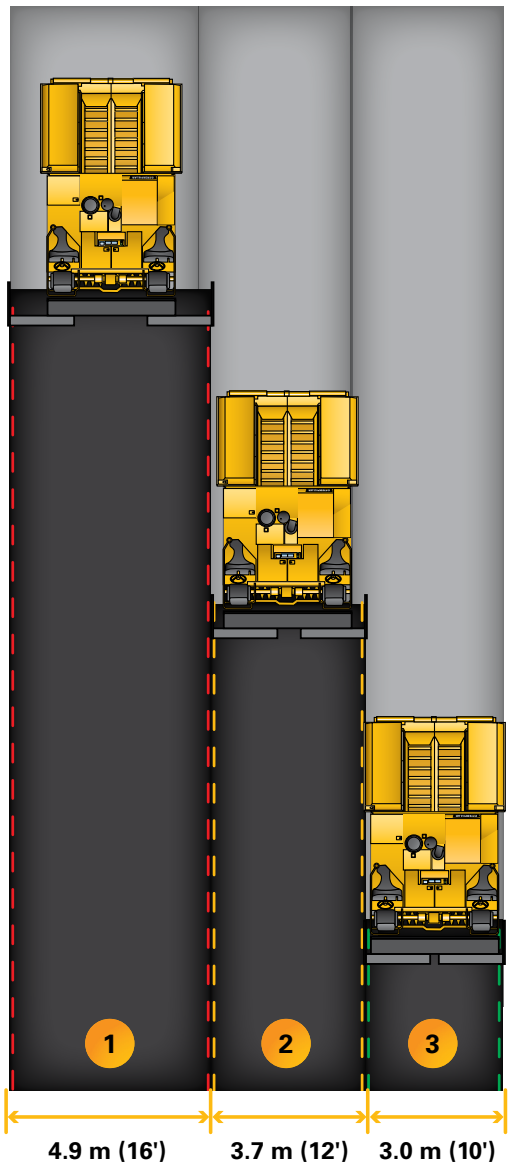
New calculations are needed for two reasons. First, the depth of paving for this layer has been reduced to 50 mm (2"). Second, the widths of two of the passes have also changed. The far left pass is now slightly wider; it's 4.9 meters (16') wide. Changing the pass width is done to avoid aligning the longitudinal joints directly above each other. The longitudinal joints should always be staggered, or offset to help prevent moisture penetration through the joints.

The center pass is still 3.7 meters (12') wide, but is now offset 15 cm (6") to the right compared to the layer below.

The right pass is now reduced to 3 m (10'). These are the final widths of the driving lanes and shoulders.

Offset longitudinal joints in an alternating fashion for each layer. The offset is normally between 15 cm to 30 cm (6" to 12"). Normally, the smaller offset should be selected if there is a slope change at the longitudinal joint. The smaller offset creates a smaller thickness change due to the slope changes.

Do the same calculations to determine the paving speed for the new layer thickness and pass widths. Assume the same hourly production rate from the plants supplying the material.



paving speed: 5 m (16.5')/min
paving depth: 50 mm (2")

SECOND LAYER

Paver 1: **4.9 m (16')**
Paver 2: **3.7 m (12')**
Paver 3: **3.0 m (10')**

Total width: **11.6 m (38')**

Now divide the total width by the width of each pass. The result will be the percentage of hourly production for each paver.

$$\text{Paver 1: } \frac{4.9\text{ m (16')}}{11.6\text{ m (38')}} = 40\%$$

$$\text{Paver 2: } \frac{3.7\text{ m (12')}}{11.6\text{ m (38')}} = 32\%$$

$$\text{Paver 3: } \frac{3.0\text{ m (10')}}{11.6\text{ m (38')}} = 26\%$$

Finally, divide the hourly production rate by the percentage of each paver. This will provide the hourly production rate in tons for each paver.

$$\text{Paver 1: } \frac{400\text{ T}}{42\%} = 168\text{ tonnes (185 tons)/hour}$$

$$\text{Paver 2: } \frac{400\text{ T}}{32\%} = 128\text{ tonnes (140 tons)/hour}$$

$$\text{Paver 3: } \frac{400\text{ T}}{26\%} = 104\text{ tonnes (115 tons)/hour}$$

The Paving Production Calculator can now be used to calculate the paving speed that will balance the three pavers on the project. Again, assume that material transfer vehicles will be utilized and the efficiency rate will be 95 percent.

PAVER ONE

Trucking	**General Inputs**	*Imperial Units*	*Metric Units*
Paver Speed	Paving Thickness:	[2.00] *in*	[50.8] *mm*
Compaction	Paving Width:	[16.00] *feet*	[4.877] *meter*
Windrow	Material Density Uncompacted:	[140] *lbs/ft³*	[2243] *kg/m³*
Yield	**Paver Speed @ Given Production Rate**		
Slope	Production Rate of Hot Plant:	[185] *tons/hr*	[168] *tonnes/hr*
Thickness	Calculated Paving Speed - 100% Efficiency:	[16.5] *ft/min*	[5.03] *m/min*
Job Summary	Calculated Paving Speed - 95% Efficiency:	[17.3] *ft/min*	[5.28] *m/min*
Legal	Calculated Paving Speed - 90% Efficiency:	[18.2] *ft/min*	[5.53] *m/min*
	Calculated Paving Speed - 85% Efficiency:	[19.0] *ft/min*	[5.78] *m/min*
	Calculated Paving Speed - 80% Efficiency:	[19.8] *ft/min*	[6.04] *m/min*
	Calculated Paving Speed - 75% Efficiency:	[20.6] *ft/min*	[6.29] *m/min*
Exit	**Effective Paving Speed**:	[16.5] *ft/min*	[5.03] *m/min*

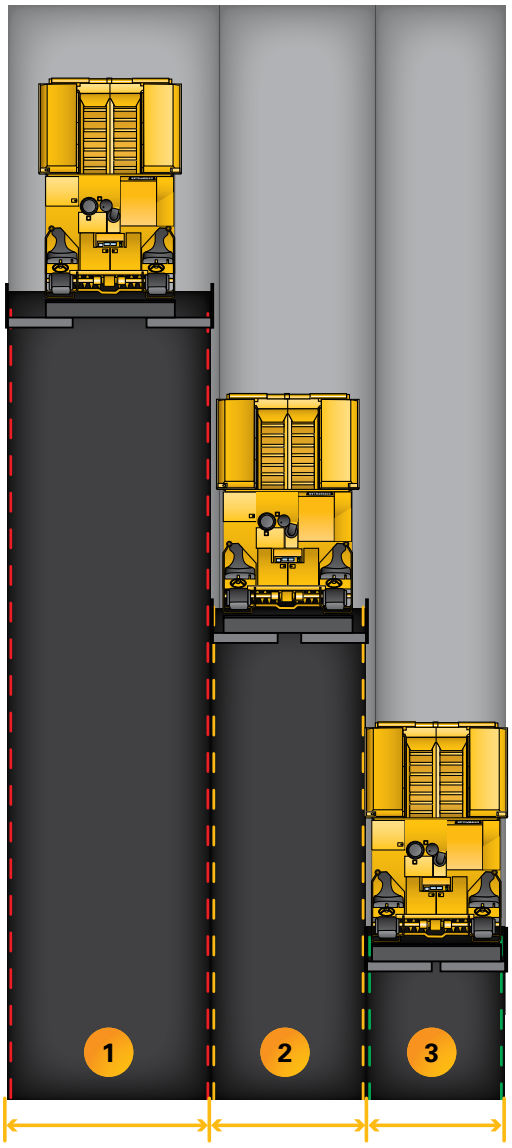
Each paver will be paving just over 5 m (16.5') per minute. There is another consideration involved with echelon paving. In some cases, the paving speeds may be affected by the density of the layer behind each paver.

THIRD LAYER

In the example shown here, there are three pavers equipped with vibratory and tamping screeds. A quality control specification on this project requires that this layer have 92 percent density behind the screed. Working at 5 m (16.5') per minute, the two pavers completing the center and right passes are meeting this specification. Working at 5 m (16.5') per minute, the paver working on the left side of the roadway is only developing 89 percent density on the bituminous layer.

The crew has adjusted the tamping frequency upward to its maximum capacity, but the screed is still not developing the required density. In this situation, there are two possible solutions. First, if another paver is available, replace the paver that is not producing the specified results. Perhaps, the tamper foot is badly worn and the screed needs repair or rebuilding. Second, slow the speed of the paver until the density reaches the required level.

The speed of the least productive paver has to be the speed of all the pavers. When paving in echelon, the pavers must stay close together.



User Tip: *When doing echelon paving, one or more pavers will be matching a longitudinal joint. Two factors need to be controlled. One is the height of the end gate ski. Be careful not to apply too much downward pressure on the ski. The ski may leave a mark in the hot, uncompacted bituminous material. Second, be sure the compactor operators have clear instructions regarding when to compact the longitudinal joint. If a trailing paver is using the area close to the joint edge as reference for the grade sensor, the compactors cannot roll the edge before the trailing paver completes the height match.*

[TRUCK EXCHANGE PROCEDURES]

Truck exchange procedures can affect the quality of the bituminous layer being placed by the crew. The drivers of the haul units should be considered part of the paving team. Drivers need training and clear instructions regarding their roles on the project.



Truck drivers must be alert for instructions while discharging material.

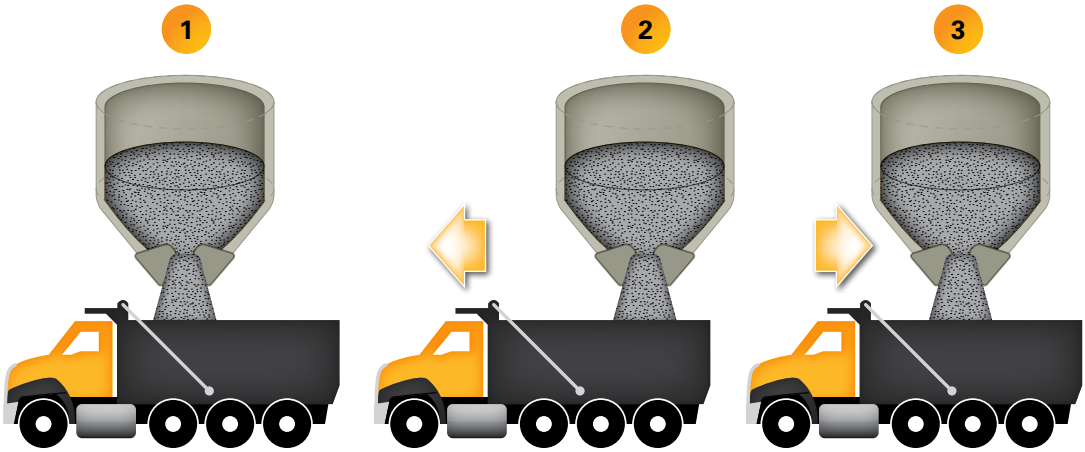
Truck Driver Responsibilities

- Coat the truck bed with approved release agent.
- Follow loading procedures.
- Cover the load.
- Drive the approved route.
- Stage where directed.
- Stop short of the paver or transfer device.
- Maintain light brake pressure.
- Watch for and follow material discharge instructions.
- Clean out where directed.

Preventing the buildup of material in the truck bed is one of the drivers' responsibilities. Most asphalt plants have a platform located near the loading area where truck drivers can apply release agent. As the film of release agent deteriorates, material begins to stick to the bed. Eventually, these agglomerations break off and contaminate the load.



Truck beds require periodic application of approved release agent.



The driver follows instructions at the plant for loading.

Truck drivers are responsible for obeying all loading instructions at the asphalt plant. There may be lights or signs that instruct the drivers to reposition trucks for different discharges from the storage silos. Once the truck beds are full, drivers receive the load tickets from the scale operator.

In most areas, the load must be covered—not just to preserve heat, but also to prevent any aggregates from spilling onto the roadway.

Each driver should receive directions for driving an approved route to the worksite. The project manager or superintendent determines the route based on traffic and load weights. It is the drivers' responsibility to stay on that route and not stop, unless there is an emergency. Trucks should arrive at the project at intervals that are consistent with loading at the plant and traffic conditions.

Depending on the type of project, haul units may have to park temporarily at a staging area. At this point, each truck driver must be alert for directions from the crew. The drivers and the crewmembers should have an agreed upon series of signals that direct the drivers' actions.

Watching the paver operator or the truck guide, the driver backs the haul unit to the front of the paver. At the signal from the operator or truck guide, the driver stops the truck just short of the push rollers. Never back into the push rollers. If the truck hits the paver, the screed will be pushed back and down and will leave a dent in the fresh layer.



The truck always stops short of the push rollers.

As the paver moves forward to contact the rear tires of the truck, the driver should apply light brake pressure to prevent the truck from rolling away from the paver. An optional hydraulic truck hitch provides a positive connection between the truck and the paver.



A truck hitch provides a positive connection between the haul unit and the paver.

As a general rule, trucks should line up in the center of the hopper to make it it easier for the paver operator to steer a straight line. However, there are times when trucks should line up off-center. For example, when paving through a superelevation (see adjacent image), the crew must compensate for a larger than normal slope to one side of the paver. The trucks should always move over close to the edge of the hopper that is aligned with the high side of the sloped surface. Gravity will move material from the trucks toward the low side of the hopper. The low side conveyor stays full while the high side conveyor runs empty. The paver operator must stop the paver to avoid running out of material in the auger chamber. When the trucks pull away, there will be a spill in front of the paver on the low side. In that situation, aligning the trucks to the high side will eliminate or minimize the spill.



Improper truck alignment leads to hopper spills.

The truck driver raises the bed to break the load and start supplying the paver hopper. The material should flow into the hopper as a surge, not as a slow trickle of material. A slow transfer of material tends to cause segregation, especially if the material has aggregates that are 19 mm (0.75") or larger.

When the bed is empty, the paver operator or truck guide signals the truck driver to pull away from the front of the paver.



Break the load and watch for signals to raise the bed.

Often, material remains on the discharge lip of the truck bed, and sometimes a little material is left in the truck bed. Clean up this material at an approved location on the project. Truck cleaning should not occur in front of the paver. The material that gets dumped on the grade when cleaning in front of the paver has to be removed. Never pave over spills on the grade or piles left from cleaning up trucks. That cleaning process takes time and interrupts production. It is always advisable to send trucks to an approved location for clean-out.



Truck clean-out should not occur in front of the paver.

When operating the paver during a truck exchange, the first option is to continue paving at normal speed while the empty truck is pulling away. The operator, perhaps with help from the truck guide, has to monitor the level of material in the hopper. As the level in the hopper decreases, the paver operator can begin to slowly raise the hopper wings to combine material from the sides of the hopper with material in the center of the hopper. The crew needs to make sure that material does not spill from the front of the hopper over the flashing while folding the hopper wings.



The paver continues to pave at normal speed as the empty truck pulls away.

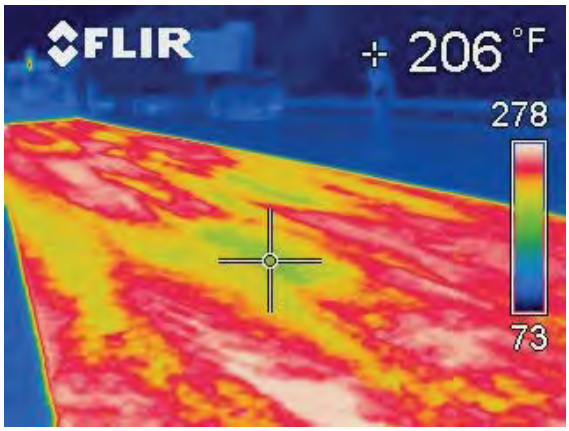
User Note: *When direct dumping from trucks into the paver, a front folding apron can help minimize spills and eliminate costly cleanup efforts.*

When the level of material has gone down in the hopper, but the conveyors are still full, stop the paver and wait for the next truck to get in position. If paving continues between trucks, there is a risk of running low on material in the hopper and in the auger chamber in front of the screed. Remember, maintaining the correct head of material in the auger chamber is a fundamental of paving. Violating that fundamental is a big mistake. The screed will begin to move downward as the head of material decreases. When the auger chamber is refilled, the screed will begin to climb. This procedure creates bumps in the mat due to inconsistent material flow.



Stop paving with the left and right conveyors still full of material.

The roughness caused by running low on material will be accentuated if the hopper wings are folded when there is little material in the hopper. Cold mix that has accumulated at the sides of the hopper will be transferred onto the conveyors without being combined with hot material. The cold material will not compact the same as hot material. Not only will the potential result be an area of low density and high air voids, the surface will also be rougher due to the compaction variation caused by the temperature differential.



Folding the hopper wings when the hopper is low on material can create cold spots in the bituminous layer.

User Tip: *Never cycle the hopper wings late in the truck exchange process when the hopper is almost empty.*

There may be times when the hopper wings should not be cycled during truck exchanges, either because of a written specification prohibiting the cycling of hopper wings or due to quality issues. For example, when placing bituminous material with large aggregates (19 mm / 0.75" or larger), there is a tendency for these large aggregates to segregate and roll to the outer edges of the hopper. When the hopper wings are folded, this segregated material tends to pass through the feeder system as a mass and show in the bituminous layer as a patch of large aggregates. The patch segregation shows up as a pattern during truck exchanges and is often referred to as end-of-load segregation. (Unit 8 will discuss segregation causes and cures.)



It is highly recommended to not cycle the hopper wings on certain projects.

In situations where the hopper wings are not to be folded, be alert for an accumulation of material at the sides and near the front of the hopper. A large amount of material can prevent the truck's tailgate from opening. The lack of fresh material can cause the conveyors to run empty, decreasing the head of material in the auger chamber. This extra material at the front of the hopper can also cause a spill directly in line with the paver's undercarriage or in line with the grade sensor if the grade sensor is located close to the side of the paver. Periodically remove this excess material from the front of the hopper to avoid spills and to facilitate the discharge of material from trucks.



Accumulations of material at the sides of the hopper can interfere with the truck's tailgate.

In many situations, the paver operator must stop the paver to wait for a new load of material. Depending on the type of mix, the weight of the screed and the length of the paver stop, the screed settlement mark may be so deep that the initial phase compactor is unable to roll out the mark. A deep settlement mark would be a sign that the operator should engage the screed counterbalance system and/or the screed lower lock.

Caterpillar does not recommend reducing the normal paving speed during truck exchanges as a means of avoiding paver stops. Remember, when the operator lowers the paving speed, the screed tends to rise. The grade control system, if used in the automatic mode, will sense the screed movement and cause the screed to move back down. A series of bumps or ripples may result from changing the paving speed to avoid running out of mix during a truck exchange.



Stopping the paver during truck exchanges can cause a screed settlement mark.

The paving speed should be calculated based on hourly production from the plant, paving width and paving depth. Calculate a speed that compensates for the paver stops during truck exchanges.

PAVER SPEED CALCULATOR

Trucking
Paver Speed
Compaction
Windrow
Yield
Slope
Thickness
Job Summary
Legal

Exit

General Inputs	*Imperial Units*	*Metric Units*
Paving Thickness:	[2.95] *in*	[75.0] *mm*
Paving Width:	[15.00] *feet*	[4.572] *meter*
Material Density Uncompacted:	[140] *lbs/ft³*	[2243] *kg/m³*
Paver Speed @ Given Production Rate		
Production Rate of Hot Plant:	[220] *tons/hr*	[200] *tonnes/hr*
Calculated Paving Speed - 100% Efficiency:	[14.2] *ft/min*	[4.33] *m/min*
Calculated Paving Speed - 95% Efficiency:	[14.9] *ft/min*	[4.55] *m/min*
Calculated Paving Speed - 90% Efficiency:	[15.6] *ft/min*	[4.76] *m/min*
Calculated Paving Speed - 85% Efficiency:	[16.3] *ft/min*	[4.98] *m/min*
Calculated Paving Speed - 80% Efficiency:	[17.0] *ft/min*	[5.20] *m/min*
Calculated Paving Speed - 75% Efficiency:	[17.8] *ft/min*	[5.41] *m/min*
Effective Paving Speed:	[14.2] *ft/min*	[4.33] *m/min*

The calculated paving speed is based on the efficiency factor that considers how material is transferred from the truck to the paver.

When the material is transferred directly from the haul unit to the paver, the efficiency rate must account for the length of the paver stops during the truck exchanges. On any project that has mostly long, uninterrupted paving passes, target at least 75 percent efficiency.

Assume that the production rate is 200 tonnes (220 tons) per hour. The paving width is 4.6 meters (15'). The paving thickness is 75 mm (3"). In that example, the actual paving speed will be about 5.5 m / minute (18' / minute) with a 75% efficiency factor. As the paving efficiency increases, the actual paving speed decreases until 100 percent efficiency is reached, or non-stop paving. In this example, continuous paving at 4.3 m / minute (14' / minute) would consume 200 tonnes (220 tons) in one hour.

Continuous paving without a material transfer device is usually not possible due to the limited storage capacity of the paver hopper. Still, some calculations can be made to determine if continuous paving at normal paving speed is possible without a transfer device. Assume, for the sake of the calculation that there are 7 tonnes (7.7 tons) available in the hopper and conveyors that could be consumed before running the conveyors empty.

YIELD CALCULATOR

Trucking
Paver Speed
Compaction
Windrow
Yield
Slope
Thickness
Job Summary
Legal

Exit

General Inputs	*Imperial Units*	*Metric Units*
Paving Thickness:	[3.00] *in*	[76.2] *mm*
Paving Width:	[15.00] *feet*	[4.572] *meter*
Material Density Uncompacted:	[140] *lbs/ft³*	[2243] *kg/m³*
Truck Capacity or Total Tonnage:	[7.7] *ton*	[7.0] *tonnes*
Length of Mat at 100% Yield:	[29.33] *feet*	[9] *meter*
Actual Length of Mat produced:	[29.53] *feet*	[9] *meter*
% Yield for given truck load or tonnage:		[101]

Thickness:	[3.00] *in*	[76.2] *mm*
Length of Mat produced:	[29.53] *feet*	[9] *meters*
Width:	[15.00] *feet*	[4.572] *meters*

The yield calculation shows the paving distance for a given amount of tonnes.

The yield calculator portion of the Paving Production Calculator shows that 7 tonnes (7.7 tons) assumed to be available in the hopper would allow the paver to place 9 m (29') of material at the width and depth of this example. The calculated paving speed at 100 percent efficiency is 4.3 m / minute (14' / minute). Therefore, approximately two minutes are available before the hopper and conveyors run out of material. If the empty truck can leave and another truck be positioned in less than two minutes, theoretically the paver could continue paving at normal speed without stopping.

In most instances, the paver cannot pave continuously between trucks without running low on material. In order to pave without stopping or slowing between trucks, the paving speed will have to be reduced. This means reducing hourly production.

[MATERIAL TRANSFER VEHICLES]

Using a self-propelled material transfer vehicle to supply the paver makes it possible to pave at normal paving speed without stopping. Various types of material transfer vehicles have different storage capacities. Some material transfer vehicles include remixing technology. In other cases, remixing occurs in the hopper insert included with the transfer vehicle.

Some public works departments have written method specifications that describe the type of material transfer vehicle that must be used on designated projects. When considering the use of any material transfer device, consult local specifications and regulations.

Material transfer devices come in different shapes and sizes, but have one common feature. They totally separate the haul unit from the paver. Therefore, the paver operator is no longer dependent on the volume of material in the hopper to determine how long to pave between trucks. Each material transfer vehicle has a specified storage capacity, and a hopper insert is frequently placed in the paver. Smaller transfer vehicles have around 11 tonnes (12 tons) storage capacity. Hopper inserts vary in capacity from around 12 tonnes (13 tons) to 16 tonnes (18 tons). Based on a fully loaded small transfer vehicle and a full hopper insert, up to 27 tonnes (30 tons) could be available for continuous paving between trucks.



A material transfer vehicle does not contact the paver as it transfers material from the haul unit.

YIELD CALCULATOR

- Trucking
- Paver Speed
- Compaction
- Windrow
- **Yield**
- Slope
- Thickness
- Job Summary
- Legal

- Exit

General Inputs	Imperial Units	Metric Units
Paving Thickness:	[3.00] in	[76.2] mm
Paving Width:	[15.00] feet	[4.572] meter
Material Density Uncompacted:	[140] lbs/ft³	[2243] kg/m³
Truck Capacity or Total Tonnage:	[30.0] ton	[27.2] tonnes
Length of Mat at 100% Yield:	[114.29] feet	[35] meter
Actual Length of Mat produced:	[115] feet	[35.05] meter
% Yield for given truck load or tonnage:		[101]

Thickness:	[3.00] in	[76.2] mm
Length of Mat produced:	[115] feet	[35.05] meters
Width:	[15.00] feet	[4.572] meters

Yield calculation showing distance of paving with added storage capacity.

The added storage capacity in the material transfer vehicle and the hopper insert increase the distance that can be paved before the paver runs empty. Assume the calculated paving speed is 4.3 m / minute (14' / minute) at 100 percent efficiency.

Since the paving distance for 27 tonnes (30 tons) is 35 m (114'), the time available for paving between trucks is about eight minutes. Compare that time to the roughly two minutes available when discharging directly from the truck into the paver.



Larger material transfer vehicles have more storage capacity.

Larger material transfer vehicles carry and store more material. A material transfer vehicle with 23-tonne (25-ton) storage capacity in combination with the large hopper insert has approximately 40 tonnes (44 tons) of material available between trucks.

YIELD CALCULATOR

Trucking				
Paver Speed	**General Inputs**	*Imperial Units*		*Metric Units*
Compaction	Paving Thickness:	[3.00] *in*		[76.2] *mm*
Windrow	Paving Width:	[15.00] *feet*		[4.572] *meter*
Yield	Material Density Uncompacted:	[140] *lbs/ft³*		[2243] *kg/m³*
Slope	Truck Capacity or Total Tonnage:	[44.0] *ton*		[39.9] *tonnes*
Thickness	**Length of Mat at 100% Yield:**	[167.62] *feet*		[51] *meter*
Job Summary	Actual Length of Mat produced:	[167.00] *feet*		[50.90] *meter*
Legal	**% Yield for given truck load or tonnage:**			[100]
	Thickness:	[3.00] *in*		[76.2] *mm*
	Length of Mat produced:	[167.00] *feet*		[50.90] *meters*
Exit	**Width:**	[15.00] *feet*		[4.572] *meters*

Yield calculation showing distance of paving with even more storage capacity.

The added storage capacity provided by the larger material transfer vehicle and the hopper insert increases the distance that can be paved before the paver runs empty. Again, assume the calculated paving speed is 4.3 m / minute (14' / minute) at 100 percent efficiency. Since the paving distance for 40 tonnes (44 tons) is 51 m (167'), the time available for paving between trucks is almost 12 minutes. Compare that time to the roughly two minutes available when discharging directly from the truck into the paver or eight minutes available when paving with a smaller material transfer vehicle.

This much paving time between trucks opens up more possibilities. For example, supplying two pavers working in echelon (staggered paving) with one material transfer vehicle is possible. This application requires some coordination but experienced crews utilize this approach on a regular basis.

The other possibility is reducing the number of trucks required.

TRUCKING CALCULATOR

Trucking	**General Inputs**	*Imperial Units*	*Metric Units*
Paver Speed	Production Rate of Hot Plant:	[220] *tons/hr*	[200] *tonnes/hr*
Compaction	Multiple Silo Plants: Initial Storage	[0] *tons*	[0] *tonnes*
Windrow	Paving Hours:	[0.0] *hrs*	[0.0] *hrs*
Yield	Truck Capacity (size):	[24.3] *net tons*	[22.0] *net tonnes*
Slope	**Truck Cycle Times** *(minutes)*		
Thickness	Load Time and Ticket:	[6]	
Job Summary	Tarp:	[4]	
Legal	Haul to Job:	[40]	
	Time on Site:	[12]	
	Dump / Clean:	[10]	
	Return Haul:	[40]	
Exit	**Truck Cycle Factor** *(total time in hours)*	[1.9]	
	Number of Trucks Needed:	[17.2]	

Trucks needed to haul 200 tonnes (220 tons) per hour with 1.9-hour cycle time.

Consider these project conditions for ordering trucks. The hourly production is 200 tonnes (220 tons) per hour. The average load for a haul unit is 22 tonnes (24 tons). Six minutes have been assigned to load from the storage silo and four minutes to cover the load. It is a 40-minute haul to the project and 40-minute return trip. On the project, assume an average of 12 minutes waiting time to unload. Assume 10 minutes to discharge into the paver hopper and clean the truck bed before leaving the worksite. The total cycle time is 1.9 hours. To deliver the hourly production rate under these conditions, 18 trucks must be ordered.

TRUCKING CALCULATOR

Trucking	**General Inputs**		*Imperial Units*	*Metric Units*
Paver Speed		Production Rate of Hot Plant:	[220] *tons/hr*	[200] *tonnes/hr*
Compaction		Multiple Silo Plants: Initial Storage	[0] *tons*	[0] *tonnes*
Windrow		Paving Hours:	[0.0] *hrs*	[0.0] *hrs*
Yield		Truck Capacity (size):	[24.3] *net tons*	[22.0] *net tonnes*
Slope	**Truck Cycle Times** *(minutes)*			
Thickness		Load Time and Ticket:	[6]	
Job Summary		Tarp:	[4]	
Legal		Haul to Job:	[40]	
		Time on Site:	[6]	
Exit		Dump / Clean:	[5]	
		Return Haul:	[40]	
	Truck Cycle Factor *(total time in hours)*		[1.7]	
	Number of Trucks Needed:		[15.4]	

Trucks needed to haul 200 tonnes (220 tons) per hour with 1.7-hour cycle time.

If a material transfer vehicle is added to the project, the cycle time will change. The hourly production is still 200 tonnes (220 tons) per hour. The average load for a haul unit stays at 22 tonnes (24 tons). Six minutes can still be assigned to load from the storage silo and four minutes to cover the load. It is a 40-minute haul to the project and 40-minute return trip. On the project, because the trucks can discharge faster into the material transfer vehicle, the waiting time and discharge time are significantly reduced. Assume an average of six minutes waiting time to unload. Assume five minutes to discharge into the material transfer vehicle and clean the truck bed before leaving the worksite. The total cycle time shrinks to 1.7 hours. Only 16 trucks are needed to deliver the hourly production under these conditions.

Studies by public works departments and universities show that material blending occurs during the material transfer process. Blending is beneficial in helping to remix material that has segregated in the haul unit. All of the normal best practices must be observed to avoid mat defects, but material transfer vehicles with remixing capability produce bituminous layers that have improved visual appearance. Remixing also helps produce a layer with uniform temperature from edge to edge. It is common to measure layer surface temperature within a 10° C (15° F) range. Most notable, the temperature variations associated with truck exchanges virtually disappear when transfer devices are used on a project.



Material blending by the material transfer vehicle improves control of segregation, uniformity and layer temperature.

The crew must be aware of a few things when using material transfer devices and hopper inserts.

The same care should be taken to discharge from the haul unit into the hopper of the material transfer vehicle that would be taken when discharging into the hopper of the paver. If any spills occur, quickly clean them up. Be especially vigilant about spills that are directly in line with the paver undercarriage or the path of the grade sensors. Remember, paving over spills is never a good practice.



Crews need to avoid spills that can occur in front of material transfer vehicles.



Run the first few loads directly through the transfer vehicle when the transfer vehicle is cold.

At the start of a paving shift, do not fill the material transfer vehicle to its storage capacity, especially when the ambient temperature is under 15° C (60° F). Hot bituminous material sticks to cold metal. There is a possibility that material will begin to clog discharge areas and decrease the amount of material that is being transferred. To prevent material sticking to cold metal, run the first two or three truckloads straight through the transfer vehicle to heat metal surfaces.

When the bituminous material includes large aggregates, there can be some segregation in the storage compartment of the material transfer vehicle and/or in the hopper insert. Caterpillar recommends that the material level in the transfer device storage compartment or in the hopper insert be maintained at least one-third full, especially if the material has aggregates 19 mm (0.75") or larger. If the remaining material in storage is used, it may primarily consist of large aggregates.



Never use all the material in the hopper insert or transfer vehicle while paving.

Just as segregation can occur in the layer when the hopper wings are cycled over a hopper that is out of material, patches of segregated aggregate will also appear in the layer when a hopper insert or transfer vehicle runs low on material. For example, if the material transfer vehicle needs to be emptied to lighten the vehicle load when crossing a bridge, do not use the last of the material for paving. Instead, discharge that material into a haul unit and salvage that last-of-load material later.



Patch segregation caused by running low in the hopper insert.

[WINDROW ELEVATORS]

Windrow elevators were the first material transfer devices. They are still widely used in some areas and are favored because they provide the most economical means of creating continuous paving at a consistent speed.



High-production windrow elevator transferring material into a paver with hopper insert.

Windrow elevators attach directly to the front of the paver.Bituminous material is discharged from haul units as a windrow onto the grade in front of the paver. Some windrow elevators have combining augers that push material toward the center of the transfer conveyor. Slats on the transfer conveyor pick up material from the windrow and deposit the material on the conveyor. Material drops off the end of the conveyor into the hopper. To increase the storage capacity of the paver hopper, many crews install rubber belt flashing across the front of the hopper or they install a hopper insert.

An adjustable moldboard is located behind the combining augers and the discharge conveyor. Set the height of the moldboard 6 mm to 12 mm (0.25" to 0.50") above the grade to avoid leaving bituminous material on the grade.

The same guidelines apply for maintaining the hopper or hopper insert at least one-third full while paving. This is especially important if paving with a mix that has aggregates 19 mm (0.75") or larger.

The most common type of haul unit used for windrow paving is referred to as a "belly dump" or a "bottom dump." This type of truck can be a single or tandem unit. The discharge is from the bottom of the truck body.

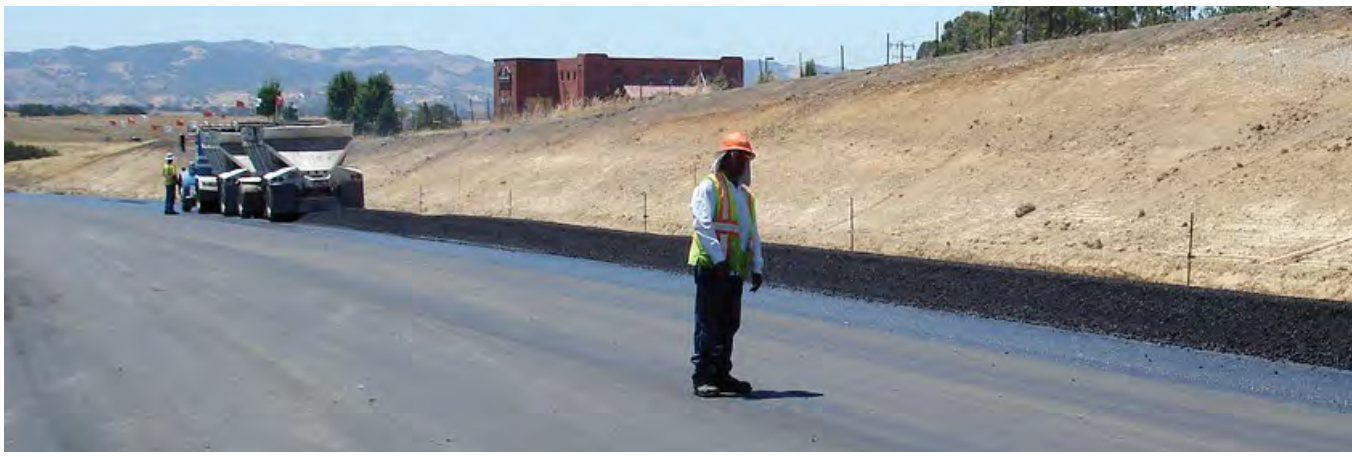
Where belly dumps are not readily available, dump trucks with modified discharge openings and live-bottom trucks have been successfully used.

The skill of the operator who is controlling the material discharge to form the windrow is critical to the paving process. This operator, referred to as the "dump man," must balance the size of the windrow with the demand for material passing through the paver. Sometimes a gap in the windrow is needed to avoid overfilling the hopper or insert. At other times, an overlap in the windrow from one truck to another is needed to supply additional material. High-production windrow elevators with wide pick-up openings make it easier to transfer large or off-center windrows.



Various types of haul units can be used to create windrows.

To avoid heat loss in the bituminous material, crews manage the length of the windrow according to ambient conditions. For example, when the ambient temperature is less than 25° C (80° F), the crew may limit the length of the windrow to what one truck can haul. On hot days, when there is little risk of heat loss, windrow length may be what two trucks can haul. Remember, every time hot bituminous material is transferred, some heat is lost. To verify heat loss, use a probe to check interior temperature of the windrow as soon as it leaves the haul unit. Then, check the interior temperature of the windrow again just before the windrow elevator picks it up. If the heat loss is more than 5° C (10° F), shorten the length of the windrow to reduce the time that it sits on grade exposed to the elements.



When ambient temperature conditions permit, there may be a long windrow in front of the paver.

[ROLLER-COMPACTED CONCRETE]

Roller-compacted concrete is a cement hydraulically-bound material used for pavements and other applications. The material is placed with asphalt paving equipment, but gains strength through compaction by rollers in addition to chemical curing that is associated with Portland cement concrete. Due to the rapid gain in strength, roller-compacted concrete may be opened to traffic much sooner than what is possible using conventional concrete pavement.

Roller-compacted concrete is a zero-slump material that is easily transported. It is most common to use end dump trucks, although mixer trucks may also deliver material to the worksite. Since it is compacted by a roller, the surface is relatively smooth and dense. It is suitable for heavy-duty, industrial applications and for road pavements where loads are heavy but speeds are low. For high-speed roadways, the surface of the roller-compacted layer is overlaid with one or more bituminous layers.

Some major applications of roller-compacted concrete include:

- Commercial parking areas
- Industrial storage and parking
- Waste transfer areas
- Container storage areas
- Truck and freight terminals
- Applications where speed of construction is essential
- Low-volume urban and rural roads or shoulders
- Aircraft parking areas (with thick bituminous overlay)



Roller-compacted concrete can be transferred to the paver by an end dump truck or by a cement mixer truck.



Thick layers of roller-compacted concrete are normally laid down with combination screeds.

Pavers perform initial compaction through vibratory screeds or through screeds that combine vibration and tamping energy. Vibratory screeds usually develop 80 percent to 85 percent density, while combination screeds develop 90 percent or more of the design density. Either way, compaction by rollers is required to reach the typical minimum density standard of up to 98 percent.

Pavers equipped with vibratory screeds normally place layers of roller-compacted concrete to a maximum depth of 15 cm (6"). If the design depth is more than 15 cm (6"), multiple layers must be paved.

Pavers equipped with vibratory and tamping screeds can develop adequate density while paving layers up to 30 cm (12") thick.